

Online Job Posts Contain Very Little Wage Information*

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Abstract

We describe the small amount of wage information in online job posts in the U.S. and how the information that is available is systematically correlated with occupation wages, firms wages, and local labor market tightness. Only 14% of posts contain any information, and 75% of all posts are accounted for by firms with pay information in less than 5% of their posts. More than half of posts with pay information list a range that, on average, is wide, spanning 28% of the midpoint (e.g. \$21-28/hr or \$32,000-\$42,000/yr). The lack of information is not ‘missing at random’. First, postings for high wage occupations and at high wage firms are less likely to provide any wage information, more likely to use ranges when they do, and wider ranges at that. Second, average wages in job postings are higher than workers’ wages in CPS and OES data, especially at the bottom of the wage distribution, by around 40 percent. In low wage salaried positions this jumps to 80 percent. Our results are exemplified by the top 20 posting private, which are representative of the U.S. economy, but have wage information in less than 2% of their posts, while the top 20 firms with wage information are unrepresentative of the U.S. economy.

Keywords: Labor, Wages, Search

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Employers post tens of millions of jobs online each year.¹ Virtually all unemployed job seekers report performing job search online.² Wage information contained in job posts is obviously of use to these job seekers. The use of wages in job posts as a substitute for administrative data on workers' wages has also proliferated in economic research. Yet, a systematic study of the availability of wages in job posts in U.S. data, and how this relates to observable differences in occupations, firms and labor markets has not, to date, been available.

Our main result is that there is very little wage information in job postings, and that the availability systematically varies across occupations and firms, but not labor markets (ranked by tightness). We study over 141 million posts scraped from over 45,000 different online sources between 2012 and 2017. The following summarizes our results:

1. **Job posts have little explicit wage information.** Only 13.5% have any wage information and just 5.8% state an exact, point wage. Private sector positions are less likely to have wage information: 10.7% with any and 4.9% with a point wage. The mean across postings disguises a skewed distribution and an even smaller median across firms. The postings-weighted median firm in the sample has any wage information in less than 1 percent of their posts. For example, 13 firms account for 4% of posts, but each have wage information in less than 0.5% of their posts. A good approximation is: there is no precise, direct, wage information.
2. **Ranges are more common than point wages.** The majority of posts with wages do not have a point wage. Instead, more than 57% feature a range. These ranges are wide on average: 28% of the mid-point. For example, \$21.00-\$27.50 per hour, or \$50,000-\$66,300 per year. This could indicate the wages of new hires are flexible, or at least designed to capture a broad set of applicants. Alternatively, there could be competitive incentives for firms to obfuscate wage information from competitors, applicants, and even current employees.

¹The data we study contains the near universe of 141,816,864 job posts in the United States for the period 2012-2017, more than 20 million a year.

²Faberman et al. (2016) show this number rose from 25% to 76% from 2000 to 2011. A Pew Research Survey from 2015 finds online search was the most important job search activity for a third of workers Smith (2015).

3. **Middle income jobs have more wage information overall.** Low and high wage occupations—measured using BLS OES data—have wage information in fewer than 10% of postings, while middle wage occupations have information in around 20% of postings. We provide statistics by hourly and salaried posts separately.
4. **Ranges are more common in high wage jobs.** However, wages are presented more opaquely in high wage occupations. For this group, more than 70 percent of postings with wage information present that information as a range, while this is less than 30 percent for low wage occupations. Ranges are also slightly wider for high wage occupation postings, and wider for salaried positions than hourly paid positions. For empirical analysis, point wages in job postings cannot be treated as ‘missing at random’.
5. **Wages in job postings systematically depart from survey data.** In low wage occupations, wages in job postings are 30 to 40 percent *higher* than observed in the BLS OES data, while they are 10 to 20 percent *lower* in high wage occupations. On the job wage growth would lead us to expect lower wages of new hires relative to incumbents. For low wage occupations, however, there is non-random and large positive selection into posting wage information. This is amplified by comparing online job posts for salaried positions and workers in salaried positions in the CPS.
6. **High wage employers present wage information less precisely.** Among all postings that do have wages—which is few—we find that within-occupation-year, firms that post higher wages are more likely to include a range instead of a point wage, and these ranges tend to be wider. These effects are significant, but small.
7. **Wage information is not correlated with local labor market conditions.** Consistent with [Kuhn et al. \(2021\)](#) we find that low unemployment labor markets are systematically tighter, but wage information is flat across markets.
8. **Lack of wage information is widespread in large private sector firms.** The most frequently posting firms are a broad, representative, cross-section of the U.S. economy: retail (Lowe’s, Sears, Macy’s, Dollar General), hospitality (Marriott), finan-

cial services (Wells Fargo, Accenture), health insurance (UnitedHealth Group, Blue Cross Blue Shield). Among the top 20, only the U.S. Air Force has wages in more than 1.5 percent of its postings. On the other hand, firms with high wage information is biased toward Federal and State Government departments, and firms in transportation and services that may pay piece rates.

Scarcity of wage information in job postings presents challenges to researchers looking to use job posting data as a stand-in for wage information of employed workers or even of new hires.³ First, the above facts relate systematic bias in the availability of any wage data at all. Second, when wage data is available, it systematically departs from data on workers in jobs. As discussed, there is clear selection into posted wages in low wage occupations, where posted wages exceed the wages of employed workers, even more so for salaried positions. Third, inferring something about average wages at firms is challenging. With so few postings per firm with wages, measures of the average wage at the firm level will be noisy, selected counterparts to administrative data.

Our work is related to literatures studying the availability of price information in markets. In labor markets, [Hall and Krueger \(2012\)](#) surveyed 1,300 newly hired workers in 2008. They found that 23% of workers knew precise pay prior to interview and that workers with less education were more likely to have had this information. Our estimate of precise wage information in only 5% of posts suggests either (i) workers contact employers with requests for information, (ii) an intervening stage between viewing a job posting and interview when an applicant is informed of a wage, which could be interpreted as a first stage of bargaining. Recent surveys conducted by [Caldwell et al. \(2023\)](#) provide support for (ii). This suggests that what researchers may have inferred from [Hall and Krueger \(2012\)](#) as wage posting, may be a first round of wage bargaining.

[Brenčič \(2012\)](#) provides the first analysis of wage information in job postings. Using U.S. data from “Monster.com” in 2006, she finds wages in 25% of posts, and that wages were more common in jobs where employers were less selective over skills or where skills were easier to measure. Using U.K. data from local employment agencies from 1988-1992,

³Our findings are specific to postings with wage information which contrasts with [Hershbein and Kahn \(2018\)](#) who find that overall postings are fairly representative. This agrees with our list of Top 20 posting firms in Table 3A, which is broadly representative of the U.S. economy.

she finds wages in 86% of posts. This suggests the explosion of internet job search may have paradoxically reduced the amount of wage information available.

Why would employers choose to withhold pay information? One literature models firms' choice between wage posting and bargaining, contrasting commitment and flexibility (Flinn and Mullins, 2021; Doniger, 2023; Cheremukhin and Restrepo-Echavarria, 2020). A second literature on pay transparency within firms offers a different perspective: employers may want to hide from current employees the wages that it is offering to new hires.⁴ A third literature on strategic price disclosure comes from an analysis of retail markets by Ellison and Ellison (2009). They show that easily available price comparisons made possible by the internet made consumers more price sensitive. In response, retailers strategically obfuscate price information. We think that our results would be useful to each of these literatures.

In recent work, Choi et al. (2020) use data from an online job board in Chile to study the cyclicalities of new hire wages. Just 16% of job posts in these data contain explicit wage information, which suggests that the low prevalence of wage data in online job posts is not unique to our data nor to the United States. Banfi and Villena-Roldan (2019) use the same data to conclude that workers are not simply inferring the wage from the text of the job posting. They observe the 'target' wage for the position, which the employer must lodge with the platform but can choose not to advertise. Job postings with materially similar descriptions and target wages receive much higher applications when a wage is posted.

Recent papers have utilized the data used in this paper to make inferences about how labor markets function. Some focus on wages: Hazell and Taska (2020) (wage rigidity), Hazell et al. (2022) (national wage setting). Our results suggest a challenge to such inference is (i) the overall lack of wages, (ii) bias in the availability of wages, (iii) systematic departure of the few posted wages that are available from wages in other data. Others focus on outcomes other than wages: Hershbein and Kahn (2018) and Modestino et al. (2015) (skill requirements), Braxton and Taska (2023) (technological change), Faberman

⁴Cullen (2023) provides an overview of the evidence on the impacts of pay transparency. Cullen and Pakzad-Hurson (2023) builds a theoretical model of pay transparency and validates it with an event study analysis of state-level pay transparency laws.

et al. (2016) (advertising relative to open positions). Our study complements this second set.

Overview. The rest of this paper is organized as follows. Section 1 describes our data. Section 2 describes the low availability of wage information, and the lack of precision when it is available. Section 3 describes how these features correlated with occupational earnings, firm wages and labor market tightness. Section 4 describes how wages deviate from what is found in surveys of incumbents. Section 5 ties this together by comparing the identities of firms that have the most postings to those that have the most wage information in their postings. Section 6 concludes.

1 Data

We use data from over 140 million online job postings over 2012-2017 provided by Burning Glass Technologies (Lightcast, 2013). Burning Glass Technologies (BGT) scraped, processed and encoded these data from over 45,000 different online sources including job boards, company websites, and others. The time frequency is daily.⁵ Proprietary algorithmic methods are used to identify and delete copies of the same job posting across platforms. For example, removing an entry from a post on a job board that also appears on the company’s corporate website.

We focus on the period 2012 to 2018. BGT data is available to us from 2010 to 2023. We start in 2012 because BGT data throughout this period is harmonized to occupation codes that only are used by the BLS from 2012 onwards. We end in 2017 because of massive breaks in 2018. From 2017 to 2018, (i) overall posts grow by 520%, (ii) the share of posts with wage information doubles, (iii) the share of posts with ranges jumps. This reflects major job boards adding their own imputed ranges of wages to posts, beginning in 2018.⁶

⁵An observation in the data is created when the job is posted. If the job is still online after 60 days, a new observation is created. Posting duration data is available after 2018. When examining these data we see almost all postings appear for exactly 60 days (more than 50 percent expire on day 61), so differential durations of job postings do not skew the data. This also means, however, that we do not think that job post duration in the data is meaningful where it is available.

⁶For example, Indeed (link: [May, 2018](#)) and LinkedIn (link: [February, 2018](#)).

As BGT does not identify these instances, this increases the fraction of posts coded with wages and the fraction with ranges.⁷ These wages are not of interest to us. We believe the explosion in the sheer number of posts in BGT data reflects the de-duplication algorithm failing once the same post has different (imputed) wages on different job boards. [Lafontaine et al. \(2023\)](#) and [Callaci et al. \(2023\)](#) carefully remove posts with words like “estimated” indicating imputation. [Lafontaine et al. \(2023\)](#) Table 2 reports that 0.2% of posts in 2017 had such features, which jumps to 58.2% in 2018.

Our statements about wage information in online job postings, hence the only question of representativeness that is relevant to us is whether the BGT data is representative of online job postings. The answer to this question is, yes.

One might be additionally interested in whether online job postings are representative of job openings more generally in the U.S. The answer to this question is, no. First, the jobs that employers post online may differ from those advertised by other means. [Hershbein and Kahn \(2018\)](#) provide a detailed comparison of BGT data to job openings measured in the Job Openings and Labor Turnover (JOLTS) and employment measured in OES data each from the BLS. They conclude BGT data are representative of many occupations and industries with a few exceptions such as over representing computer occupations and under representing food preparation and construction industries. Second, a single post may represent multiple positions. Walmart may have a single post for cashiers when it seeks to fill a dozen positions. For this reason we do not make statements about quantities of positions, which cannot be inferred from the data.

We remind the reader that the distribution of job posts differs from the distribution of workers currently employed. Below, Table 1 shows that job posts are significantly skewed towards higher paying occupations than the distribution of occupations of U.S. workers measured in OES data. As a result, much of our analysis will be conditional on occupation.

Our analysis uses only the following variables from BGT: date; annual salary; occupation (6 digit SOC); location (Metropolitan Statistical Area); employer name; and whether

⁷The share of posts with wage information in the form of ranges jumps from 57 percent to more than 75 percent, reflecting that imputed wages are stated as ranges.

the position is listed with an annual salary or hourly wage.⁸ BGT provide an annualized salary measure for compensation posted at higher frequencies. The procedure for doing so is exactly the same as used in BLS surveys.⁹ We use the Consumer Price Index to express wages in 2020 dollars. We exclude potentially interesting variables due to the high incidence of missing data (e.g. educational requirements).

We additionally use BLS data. First, employment and mean occupation wages at the 6-digit SOC level from the Occupational Employment and Wage Statistics (OES). Survey respondents are firms, and responses are aggregated by the BLS. Second, job opening (vacancies) data from JOLTS and aggregate unemployment data from the Current Population Survey (CPS). Third, individual level survey data from the Current Population Survey (CPS).

2 Presence of wage information

2.1 Pay information is scarce

Wage information in job posts is scarce. Table 1 provides basic summary statistics on 141.8 million job postings between 2012 and 2018. We consider all sectors (Table 1.I) and only the private sector (Table 1.II). Unless otherwise stated we refer to all sectors. Only 13.5% of all job postings include any type of wage information.

When we consider the distribution of job postings with wage information across firms, we see that this scarcity is more severe than this headline number suggests. We show this in Figure 1. Let x_j be the fraction of firm j job posts that contain any wage information. We then compute the share of all job postings accounted for by firms in bins of x_j . By construction, this figure integrates up to the overall mean of 13.5 percent. A fat right-tail of firms with wage information in more than 20 percent of their postings skews the means in Table 1.

⁸BGT uses textual algorithms to harmonize variations in employer names (e.g.: Hewlett Packard and HP would be the same employer) and to encode occupations based on job titles and the textual description. An audit by [Hershbein and Kahn \(2018\)](#) found occupation and industry codes to be 80% accurate.

⁹For example, in the OES, survey respondents are allowed to report pay at different frequencies. The BLS then applies the same approach as BGT to scale this up to an annual measure.

I. All sectors

	A. All jobs postings	B. Statistics by terciles of occupation income in OES		
	(1)	Low income occupations (2)	Mid income occupations (3)	High income occupations (4)
Fraction of jobs with any wage information	13.6%	9.7%	17.1%	11.6%
Fraction of jobs with a point wage	5.8%	6.3%	8.2%	3.7%
Fraction of jobs with wage info that post a range	57.5%	35.2%	52.3%	68.3%
Average percent width of wage: $(\bar{w} - \underline{w}) / (\underline{w} + \bar{w}) / 2$	27.5%	26.7%	25.8%	29.2%
Number of ads (million)	136.3	16.3	54.0	66.0
Fraction of all job postings	100.0%	12.0%	39.6%	48.4%
Fraction of OES employment	100.0%	33.3%	33.3%	33.3%

II. Private sector

	A. All jobs postings	B. Statistics by terciles of occupation income in OES		
	(1)	Low income occupations (2)	Mid income occupations (3)	High income occupations (4)
Fraction of jobs with any wage information	13.6%	9.7%	17.1%	11.6%
Fraction of jobs with a point wage	5.8%	6.3%	8.2%	3.7%
Fraction of jobs with wage info that post a range	57.5%	35.2%	52.3%	68.3%
Average percent width of wage: $(\bar{w} - \underline{w}) / (\underline{w} + \bar{w}) / 2$	27.5%	26.7%	25.8%	29.2%
Number of ads (million)	136.3	16.3	54.0	66.0
Fraction of all job postings	100.0%	12.0%	39.6%	48.4%
Fraction of OES employment	100.0%	33.3%	33.3%	33.3%

Table 1: Availability of pay information in online job postings (BGT data, 2012-2017)

Simple statistics that emerge from Figure 1 are as follows. The first blue circle marker in Figure 1A says that 50 percent of all job postings are accounted for by firms where $x_j = 0$, i.e. there is no wage information in any of the firm's job postings. This is not due to there being many firms with a single job post with no wage information, which we remove in the second line plotted. Panel B plots the cumulative density. More than 83 percent of firms have wage information in less than the overall average of 13.5 percent of their postings. Less than 10 percent of posts belong to firms with wage information in more than a quarter of their postings. In summary, most of the aggregate 13.5 percent of job postings with wage information accrue to a very small number of firms with wage information in more of their postings. We list these firms in Section 5.

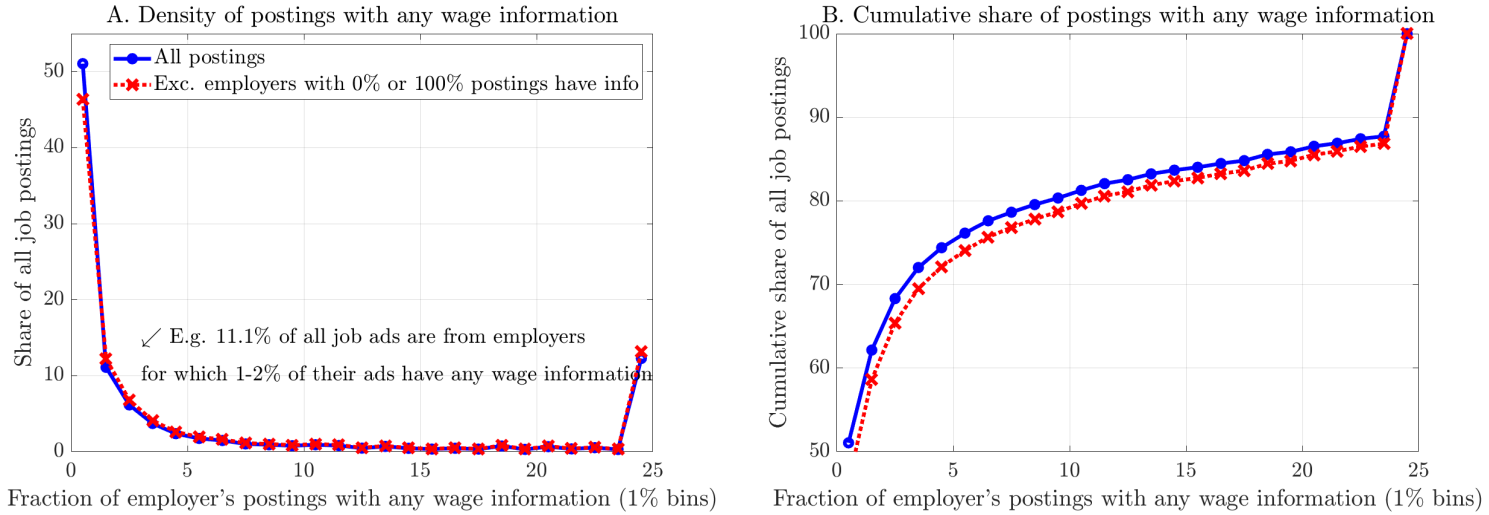


Figure 1: Distribution of wage information across firms

2.2 Available pay information tends to be imprecise

When pay information is provided, it is most often in the form of a range, that is on average wide, rather than a point wage. Table 1 shows that of the 13.5% of posts with any wage information, less than 43% of these list a single wage. Overall, 5.8% of posts have a single wage. It turns out that these ranges of wages are also, on average, quite wide. We compute the *width* of a range as the upper point minus the lower point divided by the midpoint. Across postings with a range, the average width is 27.6%. To put this in context, examples are \$21-28/hr or \$32,000-\$42,000/yr. Across sectors, private sector jobs have less information: only 10.7% have any wage information and only 4.9% have a point wage.

Summary. So far we have presented simple statistics that show that there is very little wage information available to job seekers and researchers in online job postings. However, workers and researchers may be able to fill in the blanks if wages are *missing at random*. We now show that there are systematic patterns in the availability and precision of wage information across occupations, firms and labor markets. We then show that the wages which are available deviate substantially from what one might expect from incumbent worker wages, suggesting selection in posting.

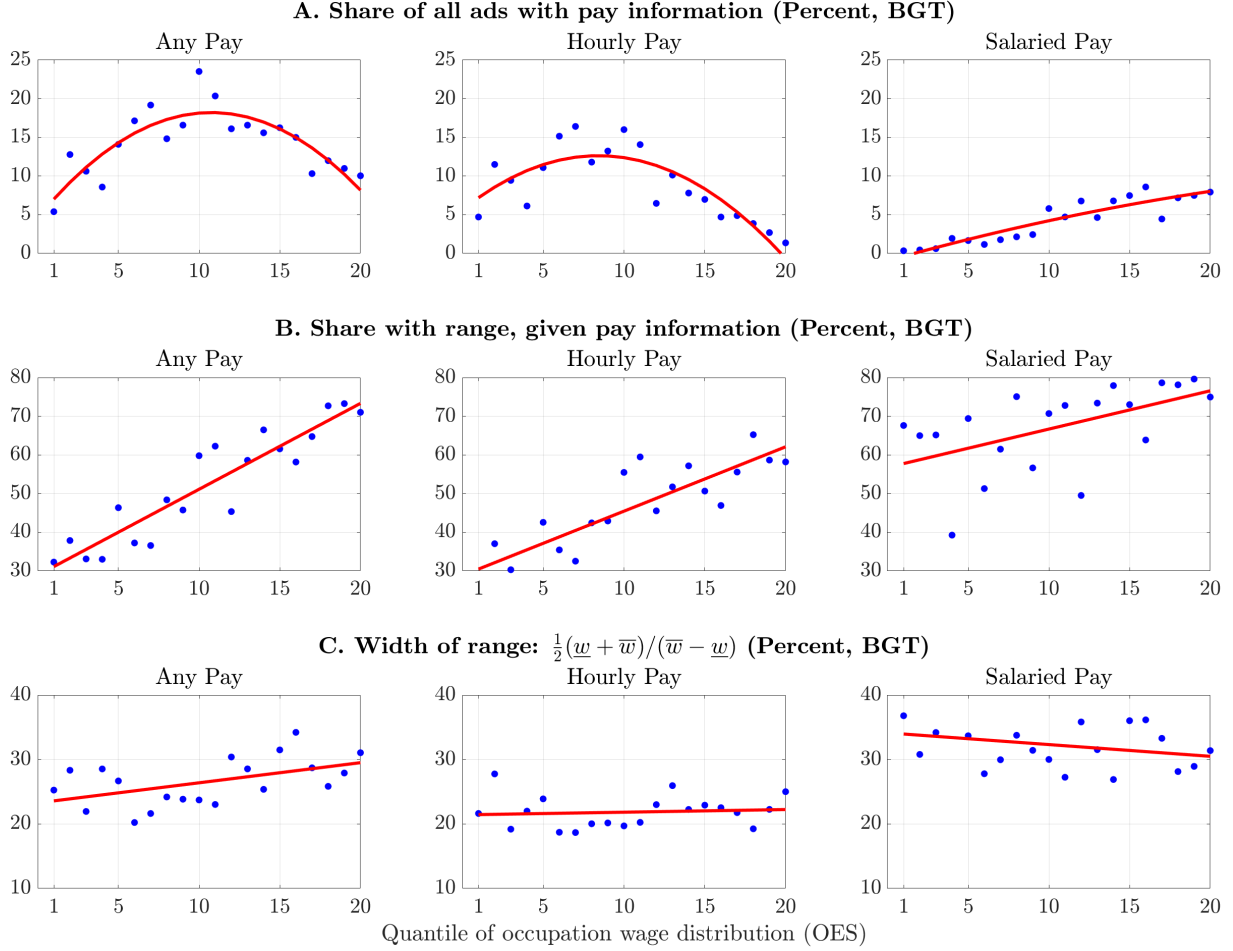


Figure 2: Comparing wage information across the wage distribution of occupations

3 Systematic variation across occupations, firms, markets

3.1 Across occupations

There are systematic differences in the availability of wage information in job posts as we move across the wage distribution of occupations. Prior research found that posts for low wage or education jobs tend to have more wage information and that low wage workers are less likely to have bargained for their wage.¹⁰ Since wage and education information is scarce in online job posts, we explore this theme using occupational wage information.

¹⁰Hall and Krueger (2012) find High School graduates were about 50% more likely than college graduates to know the pay for the job before they applied. See also Caldwell and Harmon (2019).

Occupation is coded for 96% of posts in the data.¹¹ Because wage information in job postings is scarce, we rank occupations by average earnings in the OES. Our baseline results compare workers across twenty employment weighted bins.¹²

We now begin to split jobs with wage information into those that list annual salaries and those that list hourly pay. Around 43 percent of job postings are for salaried positions. This compares to 41 percent of workers reporting salaried pay in the CPS. Overall, patterns are stronger for hourly pay, and hence these govern what we see overall. We describe these, and then turn to salaried pay.

Job postings for middle income occupations are most likely to have wage information of any type but the wage information that is available is more precise in lower paying occupations. Figure 2 shows salary information available in job postings across the occupational wage spectrum.¹³ Posts containing pay information for occupations representing the lowest paid 5% have a range only 30% of the time whereas the top 5% has a range around 70% of the time. Part of this is composition, hourly positions (column 2) are less likely to have a range than salaried positions (column 3) and the composition of job postings with wage information moves from entirely hourly to entirely salaried as we move up the occupation wage distribution. However, even conditional on salaried or hourly pay, the presence of ranges increase across occupations.

The average width of pay ranges is relatively stable across occupations, conditional on hourly or salaried pay, but much wider for salaried pay than hourly pay.

Taken together, we conclude that posts for jobs in lower paying occupations, especially positions with hourly pay, tend to have more precise pay information. Stepping back, note that nowhere in the occupation wage distribution does the prevalence of any pay information exceed 25%. Thus the conclusion that pay information is scarce in online job posts is not driven by the composition of online job posts being tilted towards higher

¹¹Table 1 column (5) shows that posts without occupation information tend to have even less wage information, hence focusing on postings with occupation information already assumes slightly more information.

¹²OES provides employment and earnings at the 6 digit level. We pool data over our sample period and employment weight to obtain a measure of the average wage at the 6 digit level. We then order occupations by the average wage and create twenty bins with equal 2012-2017 employment.

¹³The qualitative patterns we find hold when dividing the occupations into 100 wage bins (Figure A1) and when the sample is limited to only job posts from firms with more than 100 postings.

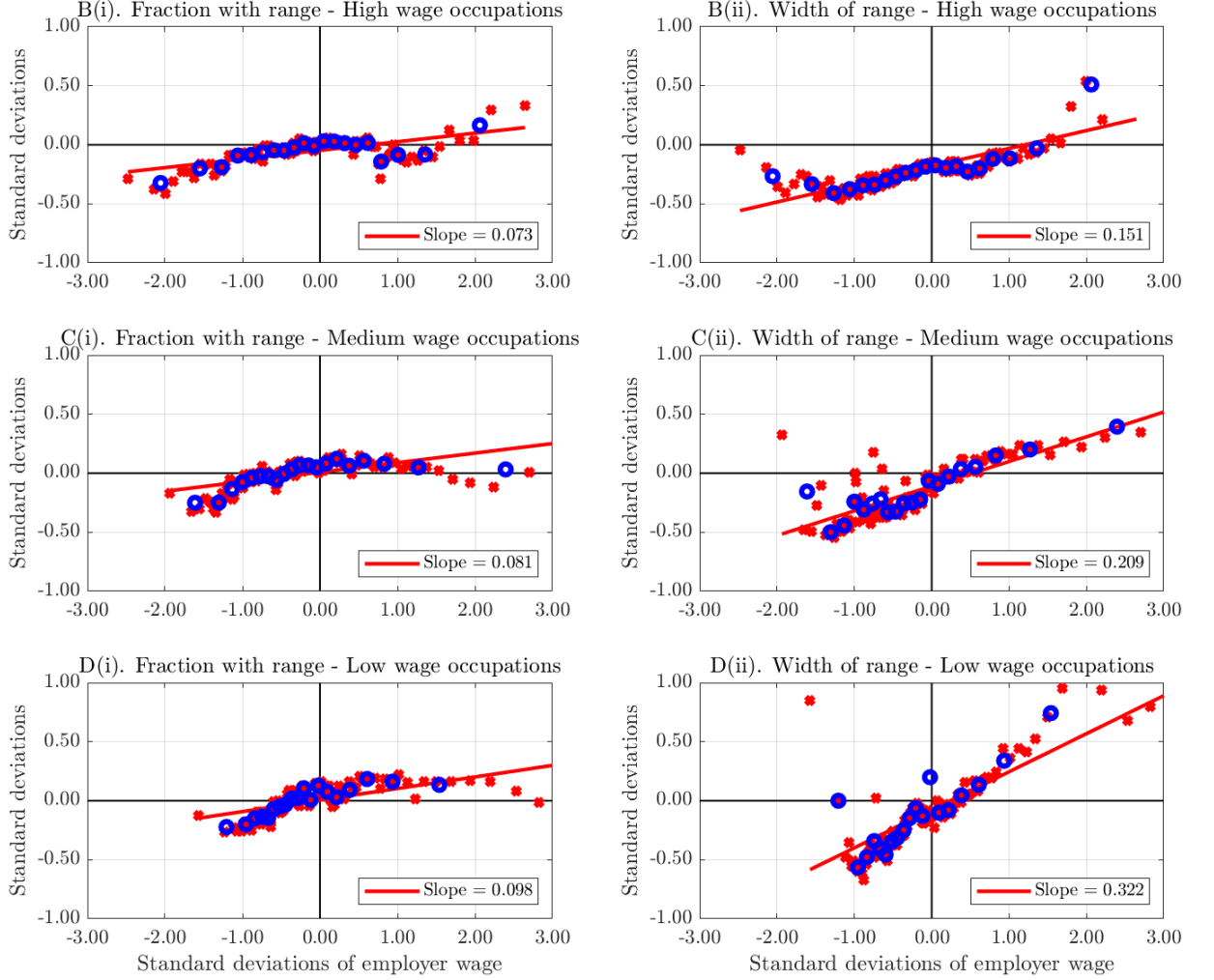


Figure 3: Within-‘occupation-wage-category’-year, Across-employer relationship between employer wage and (i) fraction of postings with a range, (ii) average width of ranges

Notes: (i) Blue circles show averages within the x - and y -variables for 20 bins of the x -variables, weighted by observations, (ii) Red crosses show averages within the x - and y -variables for 100 bins of the x -variables, weighted by observations.

paying occupations where information is scarce. This practice is broadly common across the distribution.

3.2 Across firms

In the cross-section of firms, we find a similar pattern as in the cross-section of occupations. Within an occupation category, employers with job posts with (relatively) higher

wages have a (relatively) higher tendency toward posting a range and wider percentage widths when they do.

Do employers who include more information in job posts also offer higher wages or more precise wage information? To answer this question we first drop all posts without wage or employer information and then construct a ‘firm-occupation’ as the interaction of a firm and low, middle and high wage occupations (Table 1) to acknowledge that firms may treat different wage occupations differently. Within a firm-occupation-year cell we then compute the average wage in all postings. We compute the relative wage by subtracting the within-occupation-year average and dividing by the within-occupation-year standard deviation. We repeat this for *fraction of posts with a range* and *average width of range*.

Firms with higher posted wages—conditional on occupation-year—have less precise wage information: more frequent ranges, and wider ranges. Figure 3 plots these relationships using 20 (blue circles) and 100 (red crosses) bins of relative wages. For example, from Figure 3 panels D(i) and D(ii), a two standard deviation increase in the relative wage of the firm within the low wage occupation market increases the incidence of ranges by one fifth of a standard deviation and the width of ranges by half a standard deviation. The gradient is steeper in low wage occupations.

Hence we conclude that less precise wage information is not just associated with higher paying occupations (Figure 2), it is also associated with high paying firms within occupations. In other words, not only the occupation of the worker matters but also the pay-rank of the firm. We again moderate ‘*matters*’ with the 1,000 foot view that hardly any of these postings contain wage information at all. Results in this subsection suggest that it would be difficult to characterize differences in firm wage policies using job posting data.

3.3 Across labor markets

Surprisingly, we find no systematic relationship between the availability of wage information and various measures of local labor market tightness. On the one hand, while few

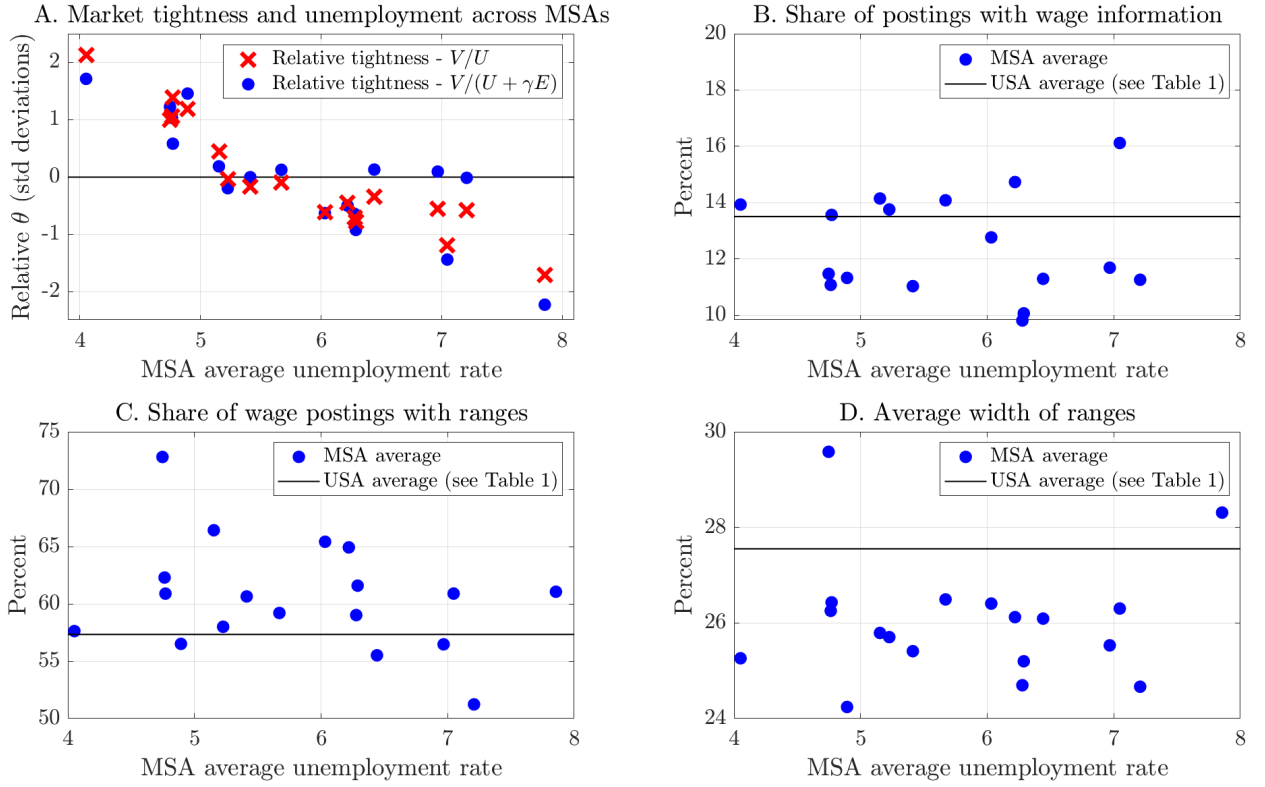


Figure 4: Wage information across labor markets

postings overall have wage information, it might be that tighter labor markets have more wage information. [Flinn and Mullins \(2021\)](#) show that under standard theories of the labor market, wage posting is more likely than wage bargaining when workers' bargaining power is stronger. A common understanding would be that workers' bargaining power is higher in tight labor markets. On the other hand, in tight labor markets employers may view the ability to negotiate after matching as a useful margin of recruiting, leading to less precise wage information in posts.

We take the lead of [Kuhn et al. \(2021\)](#) who show, geographically, that low unemployment markets are persistently tighter. Consistent with [Kuhn et al. \(2021\)](#) we first document systematic, persistent, differences across U.S. cities in labor market tightness. To measure market tightness θ we compare the number of job openings V to the number of workers searching for work S .¹⁴ Figure 4A shows that different measures of tightness

¹⁴We do not use online job postings to measure job openings, as the lack of information on the number of job openings per job posting inhibits making statements about quantities. JOLTS provides estimates of total job openings across 18 MSA's, and We measure S using either (i) total unemployed workers or (ii) total un-

both yield the expected relationship: cities with twice as high unemployment have labor markets that are slacker by four standard deviations.

Despite the tight relationship between unemployment and the prevalence of open jobs across cities, we find startlingly little relationship between measures of wage information and market tightness. The remaining panels in Figure 4 give our three measures: share of posts with pay information, share of posts with pay information with ranges, and average width of ranges. All measures are noisy around their average values taken from Table 1. This null relationship is important for researchers. It suggests little hope in disentangling various theories of frictional labor markets by appealing to the prevalence of wage posting across markets, differential in their tightness.

4 Wages in online job posts

We have documented that the availability of pay information in job postings and the precision of this information varies substantially across occupations and firms. A further question is whether there is any systematic relationship between the wage information that is available in job postings and the wages that workers are paid. Wages of new hires may be expected to be lower than those of incumbents, since the latter experience raises on the job. Surprisingly, we find that in lower paying occupations, wages in job postings are systematically *higher* than those of incumbent workers. This gap is larger when comparing salaried job posts to salaried workers (CPS) than when comparing all job posts to all workers (OES). Recall that the OES and BGT handle mappings from hourly wages to annual income in the same way.

First, Table 2 presents the mean wages in the BGT data and OES. We have two measures from the BGT data. First, we include all posts with ranges and use the mid-point. Second, we drop all posts with ranges and consider only posts with a point wage. Due to the more prevalent use of ranges in salaried positions and the higher prevalence of salaries among higher paying jobs posts, the former is higher than the latter. When we

employed workers plus a fraction 0.15 of employed workers (from estimated employed search efficiency in [Bilal et al., 2022](#)).

	A. All jobs postings	B. Statistics by terciles of occupation income in OES		
	(1)	Low income occupations (2)	Mid income occupations (3)	High income occupations (4)
BGT - includes ranges	\$62.813	\$35.339	\$47.693	\$86.632
BGT - excludes ranges	\$52.154	\$32.046	\$43.943	\$75.441
OES	\$52.589	\$25.577	\$43.346	\$96.705

Table 2: Wages in online job postings (BGT) and U.S. economy (OES)

focus on point wages, the average is similar to that obtained in the OES data (an employment weighted average across occupation-level mean wages). From the perspective of this statistic, the job postings data appears *representative* of incumbent worker wages.

Across occupations this is not the case. Online job posts have systematically higher pay in low income occupations, and systematically lower pay in high pay occupations. To reiterate, we would expect the latter as wages of new hires do not include wage growth on the job, which is reflected in the wages in the OES. Here, the surprise is what we find at the bottom of the occupation wage distribution.

Figure 5A extends this analysis to 20 quantiles of the occupation wage distribution, mirroring our analysis in Figure 2. Recall that occupations are employment weighted quantiles, defined using the OES. Each point represents the average wage in BGT data for postings in all occupations within a given quantile, divided by the employment weighted average wage in OES data for all occupations within the same quantile. For low wage occupations, wages in the BGT are around 40 percent higher than in the OES. This is large. Given the magnitude, our sense is that this reflects positive selection, that is large enough to undo the fact that online job posting wages are for new hires. That is, employers formulating job posts for jobs that will pay more find it advantageous to advertise that fact, whereas those formulating job posts for jobs that will pay less find it advantageous avoid advertising that fact.

A further data point that is consistent with this hypothesis is that the gap is even larger when focusing on salaried positions. Figure 5B narrows to job postings with annual salary information, and compares against workers that report being paid an annual salary

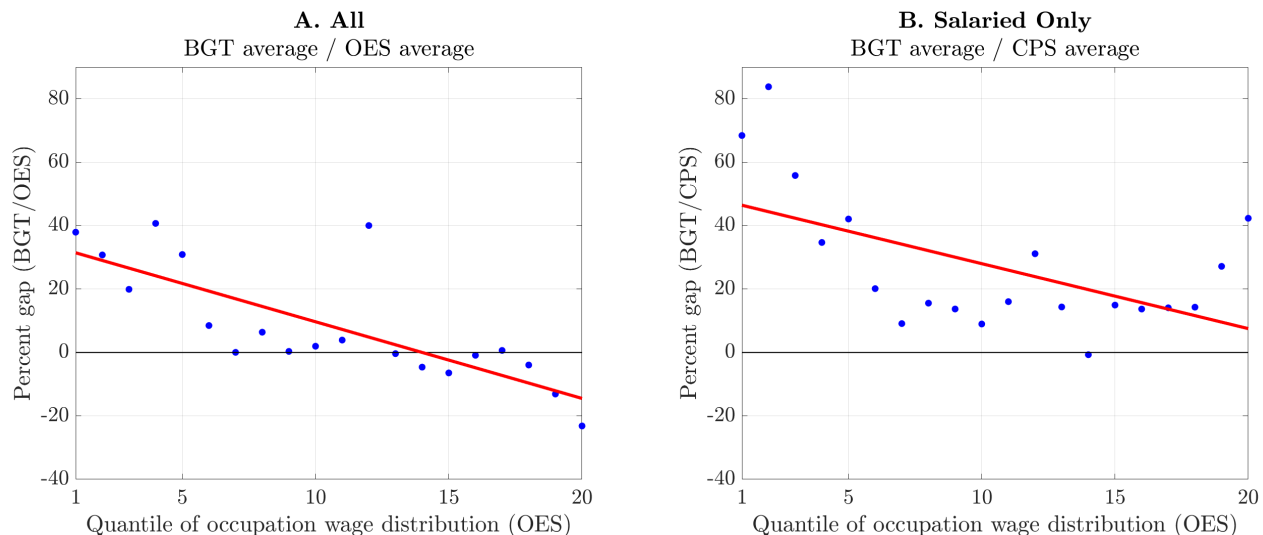


Figure 5: Ratio of average pay in BGT to average pay in survey data

in the CPS. For low wage occupations we now see salaries in online job posts that are more than 70 percent higher than in survey data. We know from Figure 2 that there are very few salaried job posts in low wage occupations. We think it is reasonable to expect that these are relatively high paying jobs, e.g. a salaried janitorial position at a large firm. Hence, the availability of wages seems to reflect positive selection in low wage occupations, which is amplified when positive selection into salaried positions is also accounted for.

5 Firms that post versus Firms that post wages

In this paper we have found that wage information in online job postings can be characterized as sparse, vague and potentially selected. To observe these attributes in action and further shed light on the information that workers have when looking for a job, we consider a simple final exercise. We compare the identities of firms that have many postings to those that have many postings with wage information. The former are representative of the U.S. private sector, but have almost no wage information. The latter are unrepresentative of the U.S. private sector, and have a small amount of wage information.

Most postings. Predominant employers in online job postings are consistent with what we would expect in the U.S. economy. Table 3A shows the employers with the most job

Firm	Total postings (1)	Share any info. (2)	Share range if info (3)
1. Lowe's Companies, Inc	768.648	0.4%	31.7%
2. Sears	670.538	2.0%	46.1%
3. Macy's	541.567	0.2%	64.3%
4. Hospital Corporation of America	524.694	0.8%	28.5%
5. Anthem Blue Cross	518.325	0.2%	25.8%
6. Wells Fargo	461.091	1.5%	16.7%
7. Dollar General	443.853	0.1%	42.1%
8. Marriott International Incorporated	441.537	1.4%	19.6%
9. The Home Depot Incorporated	407.645	0.4%	42.3%
10. Accenture	400.598	0.2%	40.8%
11. Best Buy	367.859	0.1%	23.0%
12. US Air Force	367.165	19.1%	59.7%
13. Deloitte	366.764	1.2%	4.2%
14. CVS Health	365.362	0.2%	45.0%
15. JP Morgan Chase Company	346.337	0.2%	26.9%
16. UnitedHealth Group	337.378	0.9%	34.7%
17. Bank of America	333.762	0.3%	33.5%
18. CACI	333.270	0.1%	8.6%
19. Pizza Hut	306.177	0.2%	43.3%
20. Hy-Vee	288.234	0.0%	9.8%

A. Top 20 firms by *Total postings*

Firm	Total postings (1)	Share any info. (2)	Share range if info (3)
1. Department of Veterans Affairs	266.446	62.8%	96.7%
2. US Army	220.536	45.1%	66.1%
3. Roehl Transport	138.825	51.4%	10.4%
4. US Air Force	367.165	19.1%	59.7%
5. Schneider National Incorporated	81.217	75.2%	6.2%
6. Army National Guard	135.136	38.2%	99.4%
7. Centerline	53.140	89.7%	92.3%
8. G4S	77.737	57.3%	23.1%
9. Werner Enterprises	86.697	41.6%	94.3%
10. Intermountain Healthcare	68.370	52.4%	1.2%
11. Enterprise Rent-A-Car	114.984	29.8%	4.2%
12. US Department of Defense	48.820	69.4%	94.5%
13. State of Louisiana	41.913	79.7%	72.8%
14. J.B. Hunt Transport, Inc.	66.123	48.7%	18.6%
15. Bridgestone / Firestone	146.664	21.7%	94.9%
16. Mayo Foundation for Medical Education and Research	77.732	40.7%	0.9%
17. Onward Health	36.447	85.3%	99.9%
18. YMCA	69.775	42.7%	66.7%
19. AlliedBarton Security Services	215.526	13.1%	8.8%
20. Sc Department Public Safety	31.995	86.5%	95.1%

B. Top 20 firms by *Total postings with any wage information*

Firm	Total postings (1)	Share any info. (2)	Share range if info (3)
1. Home'n'Happy	1.615	100.0%	100.0%
2. The Army Civilian Service	2.017	100.0%	94.4%
3. Rehabilitation Correction	1.075	100.0%	98.9%
4. Summerford Truck Line Inc	1.179	100.0%	100.0%
5. CRST Atlanta Expedited	1.309	100.0%	0.0%
6. Study Smart Tutors	1.503	99.9%	0.0%
7. California Department Social Services	1.209	99.9%	99.9%
8. Maricopa County Attorneys Office	1.611	99.9%	92.0%
9. Symmetry Business Group	2.818	99.8%	95.2%
10. Human Services Department	1.847	99.8%	99.7%
11. Topgear Transportation	1.189	99.7%	99.7%
12. New Mexico Behavioral Health Institute	1.745	99.7%	99.9%
13. Natural Resources Department	1.556	99.7%	93.1%
14. Minnesota State Colleges and Universities	1.451	99.7%	99.7%
15. Riverside County Regional Medical Center	2.007	99.7%	96.5%
16. North Carolina Department Of Health And Human Services	3.457	99.6%	96.4%
17. J Iverson Riddle Developmental Center	6.201	99.6%	97.4%
18. California Division of Correctional Rehabilitation	12.952	99.5%	96.4%
19. North Carolina Department Of Justice	6.991	99.5%	97.2%
20. Office Employment Training	2.083	99.5%	99.7%

C. Top 20 firms by *Share of postings with any information (firms with more than 1,000 postings)*

Table 3: Ranking of top 20 firms by different metrics, pooled over 2012-2017

posts, regardless of whether they contain salary information, from 2012-2017. These are mostly Fortune 500 companies with a broad national presence. Among the top 20 em-

employers, only one—the U.S. Air Force—exceeds the national average in terms of the share of postings with any wage information. Consistent with Figure 3A, wage information is contained in less than 1 percent of postings at 15 of the top 20 posting firms.

Most wage information. If a worker is limiting themselves to applying to jobs that feature wage information, the set of firms they would apply to would be quite different to the U.S. economy. Table 3B shows the top 20 firms in terms of total postings with wage information. Four sectors account for 17 slots: (i) Federal, State and Local government (Dept. Veterans Affairs, US Army, US Airforce, Army National Guard, US Dept. Defense, State of LA, SC Dept. Public Safety), (ii) logistics / trucking (Roehl, J.B. Hunt, Schneider, Centerline, Enterprise), (iii) security services (G4S, Allied Barton), (iv) health (Mayo Foundation, Intermountain Healthcare, Onward Health). In government, laws or other institutional guidelines often mandate wage posting. Outside government, piece-rate work (e.g. truck driving) and part-time work (e.g. security subcontracting) appear to be associated with more wage information. Similar industries are observed if we rank all firms with more than 1,000 postings by the prevalence of wage information in their posts (Table 3C). Virtually all include a wage range rather than a point wage (Column 3). Firms that have a lot of information have relatively imprecise information.

6 Conclusion

Wage information in job postings is scarce, both to job searchers and researchers. This finding is robust. Lack of wage information is pervasive in job posts across occupations, sectors, and locations. Nonetheless, there are systematic differences across these in terms of the availability of point wages, and the imprecision of ranges of wages when point wages are not given. High wage occupations and high wage firms within an occupation seem to desire more flexibility conditional on posting wage information. In low wage occupations, the wages in job postings are much higher than incumbent workers, suggesting a willingness of high paying positions to divulge this information, while low paying positions do not.

The systematic availability of wage information in job postings raises concerns about research taking wages in online job posts as a proxy for wage offers to new hires or wages paid to continuing employees. There are some recognized advantages of these data over administrative measures of wages. They are high frequency, at the firm and establishment level, spatially precise, and contain occupation data. Yet as we have seen, the most common outcome is that wage information is missing, and the incidence of this is selected across occupations, sectors and firms. One might use the mid-point of postings with ranges to expand the data available. However, (i) the wide width of ranges is a cause for concern, (ii) our analysis shows that the choice of posting a range is also not random, and hence dropping posts with ranges also introduces bias. Wide, variable ranges are a problem that remains salient even after the enactment of recent laws mandating wage information in job posting on top of substantial non-compliance ([Arnold et al., 2025](#)).

One might be hopeful that recent and proposed laws mandating wage information in job posts would facilitate imputation. We are skeptical. Our preliminary analysis of Colorado after it's wage transparency laws were enacted in shows compliance is far from complete.¹⁵ The incidence of any wage information has increased, but via more frequent and wider ranges. Posting a range satisfies the law, but keeps wages opaque for workers and researchers.

Our findings provide fertile ground for researchers trying to answer open questions in labor markets. First, why do employers choose to post little wage information? There is a growing body of work in this area ([Cheremukhin and Restrepo-Echavarria, 2020](#); [Flinn and Mullins, 2021](#); [Doniger, 2023](#)). Theories proposed should be consistent with the facts presented here: many of the top employers present essentially zero wage information, there is systematic variation across occupations, and no variation across locations. As new theories are developed, one can return to our results to test implications. Second, does scarce wage information inhibit matching or dissuade search? Some research suggests that workers get wage information elsewhere, whether in additional text in the job post or through social networks. Still, employers are systematically making the choice to

¹⁵Preliminary work by [Arnold et al. \(2022\)](#) estimates that this law increased the number of job postings with wage information by 25 percentage points.

withhold wages. Third, does the lack wage information in job posts indicate a higher than previously thought incidence of bargaining? Detailed surveys on the stages of bargaining like [Caldwell et al. \(2023\)](#) will be useful.¹⁶

¹⁶Results from this work are consistent with our view that bargaining is wide spread. Consider a worker filling out a job application that did not state a point wage (95% of posts), where the application asks for a expected salary range, and the firm replies with a wage ahead of the interview. Answering the survey of [Hall and Krueger \(2012\)](#), the individual *Did know the precise wage when interviewed*, but the process represents bargaining. [Caldwell et al. \(2023\)](#) finds that the vast majority of application processes elicit a candidate's wage expectations. [Agan et al. \(2021\)](#) finds that employee responses to salary expectations questions are interpreted by firms as indicative of outside options and worker productivity.

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ONLINE APPENDIX - Additional figures

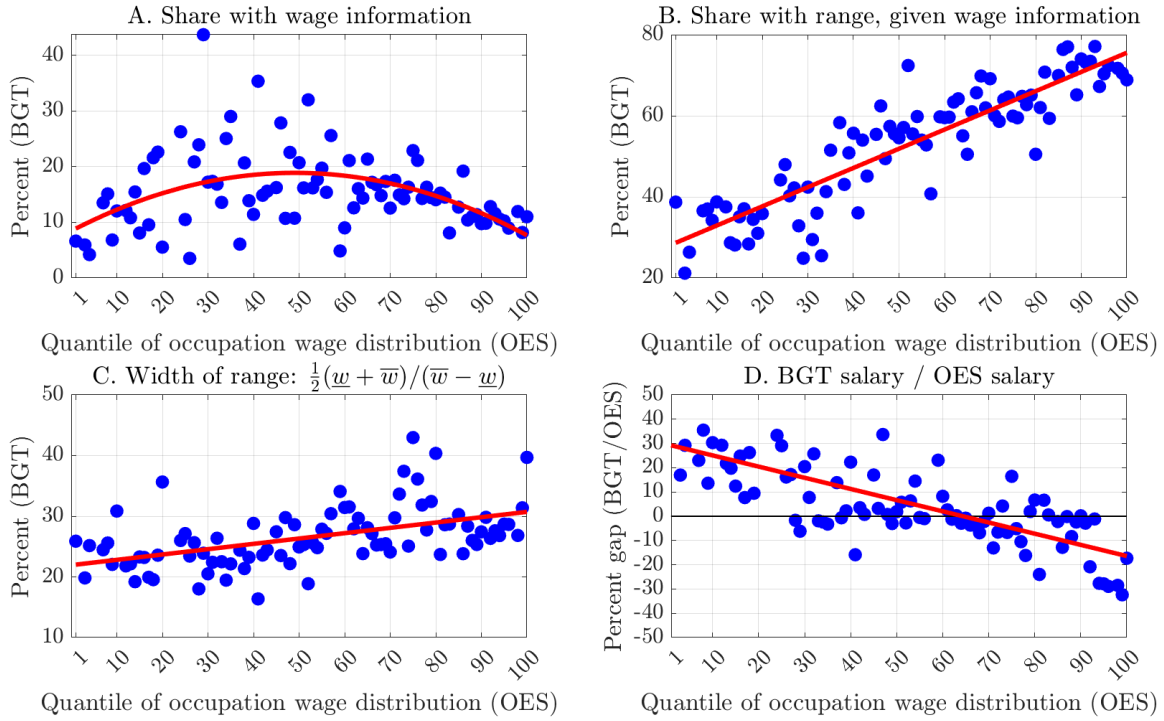


Figure A1: Within-‘occupation-wage-category’-year, Across-employer relationship between employer wage and (i) fraction of postings with a range, (ii) average width of ranges